Supplementary Material: Content-Conditioned Cyclic Seam Carving for Equirectangular Panoramas

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***Abstract*—This supplement holds extended protocol, figures, and analysis removed from the main paper to meet the page limit. Locked experimental numbers in the main text are not changed here. Code and remaining data will be attached with this file.**

***Index Terms*—equirectangular panorama, seam carving, supplementary material**

I. Code and Remaining Data

The following will be included with this supplement (not yet packed in this PDF):

- Source code for cyclic seam routing, spherical energy $E_p = E_g \cos\phi$, the content-conditioned gate ($\tau = 0.20$, $k = 24$), and the SegFormer-B0 ADE20K person-map wrapper.
- Configuration files and scripts that reproduce Tables I–III of the main paper at width at most 512.
- Remaining evaluation outputs (per-image hit rates, gate ON/OFF flags, temporal photometric residuals) that do not fit in the main text.

PanoCrowd/PanoCount stills and head-point annotations follow the collection cited in the main paper [1]. That counting paper is accepted at PRICAI 2026; the implementation is public at https://github.com/aiguo112/PanoCount360. The stills used in the carving experiments are not treated as a public downloadable benchmark; dataset access follows the request process of that repository.

II. Extended Experimental Protocol

Polar band means the top and bottom 15% of rows.

Occupancy. Fraction of seam pixels whose row lies in the polar band. Any vertical seam occupies that band at a rate fixed by the band height (here 0.297). Occupancy is a sanity check, not the spherical outcome.

Routing share. Fraction of cyclic $|\Delta x|$ whose starting row is in the polar band. A wrap step counts as $1\,\mathrm{px}$ using the seam's own width. Mean polar $|\Delta x|$ is the mean cyclic step on polar-band rows, in pixels.

Head-hit rate. A GT head is hit if any of the first k original-width seams, on that head's row, satisfies cyclic $|x - x_{\mathrm{head}}| \le r(\phi)$, with

$$r(\phi) = \frac{r_{\mathrm{eq}}}{\max\big(|\cos\phi|, \cos 75^\circ\big)}, \quad r_{\mathrm{eq}} = W \cdot \frac{2^\circ}{360}. \tag{1}$$

Reported as a percentage of GT heads.

Body-hit rate. A GT person is hit if any of the first k original-width seams cuts that person's constructed body column: a meridian strip from the head downward by 12° of latitude, half-width $2.5\,r(\phi)$. Reported as a percentage of GT persons. Body-hit is not head-hit and is not a segmentation IoU.

Predicted-person instance hit rate. A SegFormer-B0 [2] model trained on ADE20K [3] (`nvidia/segformer-b0-finetuned-ade-512-512`) predicts a per-pixel class map. The person class is selected by label name; there is no logit threshold beyond argmax. Instances are eight-connected components of that binary map with no minimum size. Components are computed on the planar $H \times W$ grid and do not wrap across $x = 0$ and $x = W - 1$. The metric is the fraction of those instances touched by any of the first k original-width (b) seams.

Gate. Let h_{pred} be that fraction for $k = 24$. If there are no predicted-person instances, $h_{\mathrm{pred}} = 0$ and the gate is off. When the gate is on, (c) is used ($\lambda_s = 4$ on the predicted $0/1$ person map); when it is off, (b) is kept. Both seam sets are always computed from the original image; the gate only selects which set is the method output. This instantiation is not equivalent to the oracle that paints GT body columns.

Temporal photometric validation. Equatorial phase correlation [4] estimates a cyclic horizontal pan; the polar 15% of rows is excluded. Photometric residual is measured on the same equatorial band, before and after warping I_t by the estimated pan. Post-warp photometry, not pan magnitude, is the test of whether the estimate reflects camera motion.

Statistics. The 95% CI in the main paper is the paired normal-approximation interval on $n = 171$ images. Many per-image differences are exactly zero, including all gate-off images. The mean is the paired average on the pre-specified medium split; it is not a typical per-image effect, and no multiple-testing correction is applied. The practical SegFormer run was executed after the oracle GT analysis and is not a pre-registered hypothesis test of the gate.

III. RQ1 Scatter

On CVRG-Pano val+test ($n = 120$), the pre-registered content predictor—unweighted polar/equator Sobel mean, range 0.14–0.74—does not track polar routing Δ (Fig. 1). Every image has polar Sobel mean below the equatorial mean (ratio < 1). Spherical energy therefore remains a legal term. On this

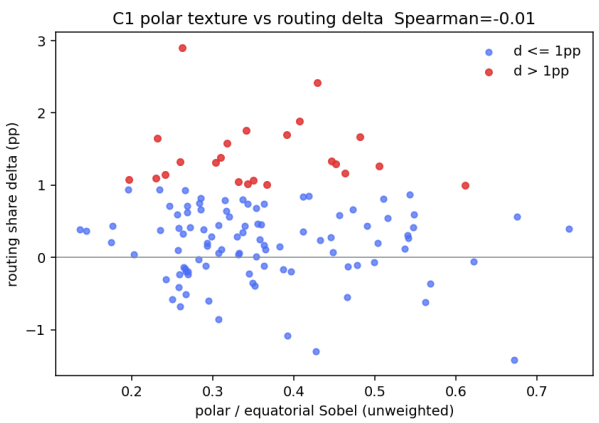


Fig. 1. Unweighted polar/equator Sobel ratio versus polar routing Δ on CVRG-Pano ($n = 120$). Spearman $\rho = -0.01$. Red points: $\Delta > +1$ pp. The > 1 pp tail is not the high-texture side of the predictor. The predictor is not a successful account of the routing change.



(b) spherical, largest medium Δ (c) 2° head-disk energy

Fig. 2. PanoCrowd medium example with the largest GT head-hit reduction at $k = 24$ under always-on 2° head-disk energy. This is not a typical image: $87/171$ medium frames are $0 \to 0$. Red: spherical baseline (b). Green: (c) seams. Even here the baseline already misses most 2° heads.

corpus it is a small routing effect, not a large polar detour, and it is not explained by polar texture.

IV. Head-Disk Qualitative Example

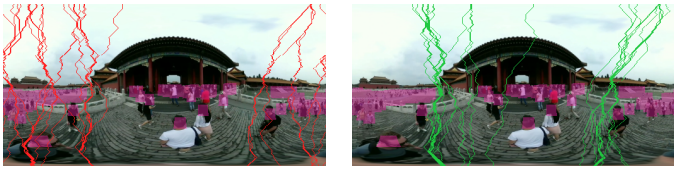
Among the 84 medium images where (b) hits at least one head, $82/84$ go down under always-on head-disk energy. The other 87 are $0 \to 0$ ties. Scene composition—thin seams versus clustered 2° disks—leaves little head-cut to undo.

V. Oracle Body-Column Example

On the oracle high-body subset, (c) uses essentially all of the baseline damage. On the complement, (b) already hits 3.4% of columns. The all-medium oracle mean is the 49-image tail. Body columns remain a constructed proxy (near-field boxes are mostly head and shoulders; dense crowds merge).

VI. Why Conditional Protection?

The practical gate changes the seam set on $56/171$ medium images. The other 115 keep the spherical baseline. The all-medium -1.55 pp is therefore a mixture: a nonzero paired change on the selected subset (-4.73 pp) and a structural zero on the rest. On the selected subset the GT sign is mixed (34 down, 5 tied, 17 up), so the ON mean is not a claim that every activation helps on GT columns. The experiment describes where λ_s has work to do. It does not show that $\tau = 0.20$ is optimal, or that the gate is better than other simple rules a practitioner might try after seeing the same regime split. Gain is concentrated where predicted baseline risk is at least τ, which is the regime the rule was designed to isolate. That pattern is consistent with semantic protection being regime-dependent on this split. It is not a causal claim that $\tau = 0.20$ is optimal, nor that SegFormer person pixels cause the GT reduction: the gate is a selection rule, and GT body-hit is a held-out score of a different object.



(b) spherical seams (c) GT body-column energy

Fig. 3. Largest oracle high-body example. Magenta: constructed GT body columns, not SegFormer. Red: spherical baseline (b). Green: always-on (c) after $\lambda_s = 4$. Illustration of the $49/49$ direction, not a mask-quality claim.

The same concentration appears in the oracle table, at a different scale and with a different detector. High-body GT columns give -35.2 pp on 49 images and -3.4 pp on 122. The practical ON set ($n = 56$) overlaps that oracle subset on only 30 images (19 high-body images remain gate-OFF). Predicted-person instances are not GT bodies, and -4.73 pp is not -35.2 pp. Conditional use of λ_s is supported as a description of where the energy has work to do, not as equivalence of the two gates.

VII. Temporal Control

Temporal compensation is not part of the proposed carving pipeline. This section is a control on the camera-motion assumption a compensated-frame method would need.

The control fixes one real ERP frame I_t (clip ERP2, source 1626, width 512).

(A) **Known cyclic yaw.** Rolling the full ERP by $k \in \{8, 16, 32, 64, 128, 141\}$ pixels, the estimator recovers each k exactly and the post-warp photometric residual is zero. Phase correlation is correct on this projection when the motion is a genuine yaw.

(B) **True consecutive frame** ($\Delta t = 0.03$ s). Estimated pan 0.00 px; equatorial dense-flow u 0.00 px. Warping does not change the residual (ratio 1.00). At video rate there is no motion to regularize, and the estimator introduces none.

(C) **Linspace neighbor** ($\Delta t = 7.74$ s; 232 frames). The estimator returns 253 px on the worst pair (clip-mean absolute pan 141 px). Warping I_t by that shift leaves the equator essentially unchanged: residual falls only to 0.95 of raw ($56.2 \to 53.5$).

A warp that explained camera motion would drive the residual toward zero, as in (A). Condition (C) does not, so the 253 px shift does not account for the sparse neighbor. Consecutive-frame windows of ERP2, ERP4, and ERP5 in this corpus agree near zero; we found no validated video-rate yaw window. This is not a test of whether temporal seam

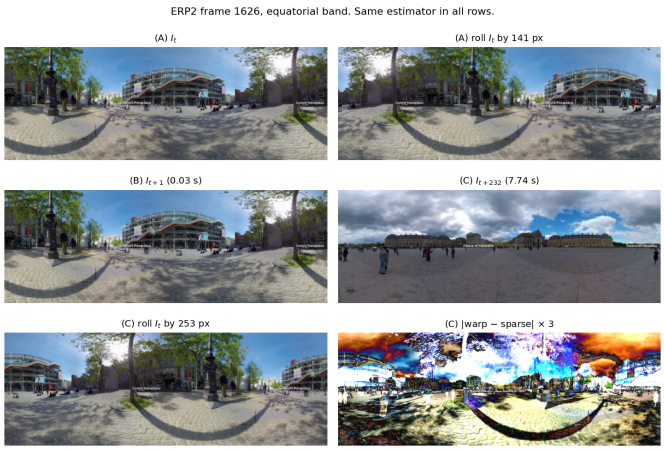


Fig. 4. Same ERP frame I_t (clip ERP2, source 1626), equatorial band. (A) A known 141 px cyclic roll is recovered exactly. (B) The next video frame is already aligned (pan 0). (C) At 7.74 s the estimator reports 253 px, but rolling I_t by that amount does not explain the sparse neighbor. Photometric residual, not pan magnitude, is the validation criterion.

propagation works when genuine inter-frame yaw is present, and it is not offered as a successful temporal method.

VIII. Qualitative Notes

In every main-paper qualitative panel the magenta overlay is SegFormer predicted person pixels, not GT. GT body-hit in the caption is the evaluation score, not the gate statistic h_{pred}.

Stem 166 is gate OFF: predicted instance hit is 0, GT body-hit is 0, and the output equals (b). Stem 242 is also gate OFF ($h_{\mathrm{pred}} = 0$) but GT body-hit is 16.9%: predicted components miss people that the GT columns still count, so the gate leaves residual GT damage untouched. Stem 800 is gate ON and a successful protection case on the GT score ($26.7\% \rightarrow 15.5\%$), one of the $34/56$ ON images where GT body-hit falls; this image is also in the oracle high-body set, which we note as overlap, not as proof that the practical detector found GT columns. Stem 915 is gate ON with the opposite GT movement ($11.7\% \rightarrow 22.3\%$), one of the $17/56$ ON images where predicted-person energy increases GT body-hit when the two person representations disagree. These frames illustrate operating regimes. They are not a random sample and are not a performance claim.

References

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